MID-TERM EXAMINATION
(B.Tech. CSE/ECE) 1st semester
(January, 2023) OFF LINE mode

Subject Code: BEC-110	Subject: Basic Electrical Engineering

Time: 1 ½Hours Maximum Marks: 30

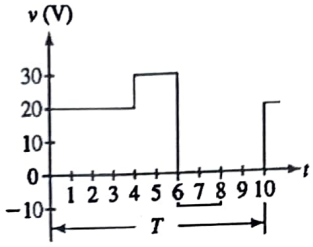
Q1 Attempt all Parts (2.5*4)

(a) State Nulling effect and prove Thévenin's theorem.

(b) If resistance of a circuit R=2.5 Ω,Inductance of circuit L=0.02 H and capacitance of circuit C =0.2 μF. Calculate-:

i. Resonance Frequency
ii. Q of a series circuit.

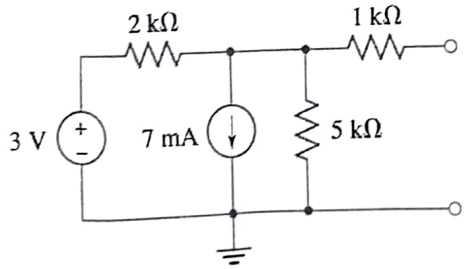
(c) What do you understand by effective value? Figure below shows the one cycle of a voltage waveform. Determine its effective value.



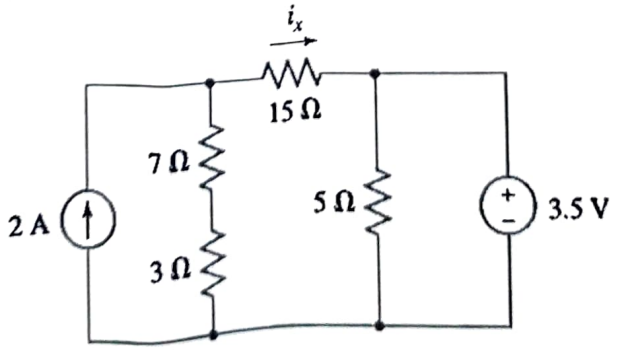
(d) A coil connected to 100V DC supply draws 10A and the same coil when connected to 100V AC voltage of frequency 50Hz draws 5A.Calculate the parameters of the coil and power factor.

Q2 (Attempt any Two Parts) (5,5)

(a) State maximum power transfer theorem. Determine the Thévenin and Nortan equivalents of the circuit shown below:



(b) State Millman's theorem. Use superposition theorem to compute the current ix for the circuit shown below:



(c) A circuit having a resistance of 4 Ω an inductance of 0.5 H and a variable capacitance in series is connected with a 100 V, 50 Hz supply.Calculate-:

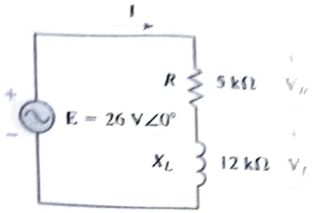
i. Capacitance to give resonance
ii. Voltage across Inductance
iii. Voltage across Capacitance
iv. Q-factor of the circuit

PTO

Q3 (Attempt any Two Parts) **(5,5)**

(a) Three resistance 1R, 2R, 3R are connected in delta.Determine the resistance for an Equivalent star.

(b) Consider the circuit shown below. Find Z_T. Determine the voltages V_R and V_L using the voltage divider rule. Verify Kirchhoff's voltage law around the closed loop.



(c) The equation of an alternating current *i* is *i*=42.42 sin628t. find (i)max value (ii)frequency (iii) R.M.S. value (iv) average value (v)form factor